Coordination-Collapse Phase Boundary in Policy-Gradient Cooperative MARL

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Abstract

Cooperative multi-agent reinforcement learning (MARL) is known to collapse into a degenerate *lazy-agent* policy when reward centralization falls too low, but this failure has never been characterized as a quantitative object. We treat it as one: we measure the critical centralization w_{crit} at which coordination gives way to collapse—with a pre-registered collapse definition and bootstrap confidence intervals—turning a qualitative failure mode into a measurable phase boundary, the first in deep cooperative MARL. The boundary is an instrument, and our central finding uses it to characterize how *loss aversion* governs coordination: loss-averse agents ($\lambda > 1$) systematically *raise* w_{crit}—they over-weight the immediate local cost of cooperating against its diffuse, delayed global benefit, so the team needs more reward centralization to coordinate. This is our most strongly supported result: the rightward shift is monotone in λ, saturating, and *replicates across four environment–algorithm settings*: the supply chain under PPO and under centralized-critic MAPPO, MPE Cooperative Navigation under MAPPO, and Level-Based Foraging under MAPPO—i.e. three environments and two algorithms (four of the six possible combinations). On the supply chain a full sweep λ=1…7 is tracked by a one-parameter, theory-bounded law (RMSE 0.014): the shift is $+0.089$ $[0.041, 0.141]$ at λ=3 (within the human range) and up to $+0.166$ $[0.135, 0.202]$ at λ=7 (a high-stakes upper bound). A mean-field signal-to-noise account *fixes this sign a priori* (proved as a sign invariant of the escape barrier) and is independently corroborated by an exploration-noise ablation; both the metastable horizon dependence and the exploration-noise effect further *generalize* beyond the supply chain (metastability to MPE and LBF; the noise effect to LBF, with a mechanism-predicted null on MPE). Two further results sharpen the picture but are scoped to the supply-chain substrate, where they hold cleanly. A three-way decomposition shows probability-weighting curvature moving w_{crit} in the *opposite* direction (left by 0.100 $[0.033, 0.172]$); and—a striking, falsifiable prediction we establish on this one substrate—the boundary is set by a *minority by network position*: a single *downstream* loss-averse node drives $w_{\mathrm{crit}} = 0.853$, indistinguishable from a fully loss-averse population (0.851), whereas the same bias placed upstream moves it far less (0.726). Where a directed resource-flow ordering exists, this predicts a system's centralization requirement is fixed by its most downstream biased participant, not the mean. The boundary doubles as a safety margin: how much more centralization a cooperative system needs once its agents are loss-averse.

Introduction

Cooperative systems increasingly mix human decision-makers with learning agents trained to act like them. In automated replenishment, fraud review, energy dispatch, and clinical triage, the deployed agents do not optimise a clean objective—they inherit human preference structure by design, either because they are trained on human choices through preference-aligned objectives that are themselves prospect-theoretic in form (Ethayarajh et al., 2024), or because they act alongside humans in the loop (Lalmohammed et al., 2025). The most robustly documented feature of that preference structure is *loss aversion*: people—and now the agents shaped after them—weigh a loss more heavily than an equivalent gain (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992). Individual-level behavioral economics tells us what this does to one decision-maker. It does not tell us what it does to the *coordination* of many decision-makers learning together (Leibo et al., 2017; Lowe et al., 2017)—which is the question a deployer actually faces.

That coordination has a well-documented way of failing. Cooperative multi-agent reinforcement learning (MARL) collapses into a degenerate, zero-action *lazy-agent* policy when agents abandon cooperation (Liu et al., 2023; Shi and Peng, 2023). Whether a system collapses depends on its *reward centralization* $w \in [0, 1]$—the fraction of each agent's reward drawn from the team's global outcome rather than its own local one: push w too low and each agent's own performance drowns out its contribution to the team, so cooperation carries no learnable signal and the safe, lazy policy wins. Yet the field describes this dependence only qualitatively—"centralization helps" (Akella, 2025; Liu et al., 2023). *Where*, in w, coordination gives way to collapse—the critical centralization w_{crit} at which the collapse probability crosses one-half—has never been measured. It is a slogan, not a quantity.

We ground the measurement in automated replenishment for multi-echelon supply chains, and the choice is not incidental: it is the kind of domain where the sharpest version of our question can even be *asked*. A supply chain has a *directed flow*—inventory moves downstream along a chain, so "upstream" and "downstream" are well defined—which lets us ask not merely whether a behavioral bias matters but *where in the network* a biased participant matters most. Its failure modes are economically legible (hoarding, stockouts, the

bullwhip cascade), so collapse is a recognisable operational disaster rather than an abstract training pathology; and real replenishment is done by planners who demonstrably weigh losses more heavily than gains, so the agents replacing or assisting them inherit that bias by construction. The substrate thus supplies the directed structure, the legible failure, and the behavioral mechanism in one place—which is why our most fine-grained results live here, and why we are explicit about which of them we claim beyond it.

We treat collapse the way physics treats a phase boundary: a pre-registered, frozen binary collapse definition, logistic estimation of $P(\text{collapse} \mid w)$, and bootstrap confidence intervals on w_{crit}. We use "phase boundary" in the empirical sense—a sharp, metastable threshold whose asymptotic ($T \to \infty$) value is the boundary proper—not as a claim of a thermodynamic-limit singularity. The boundary is then an *instrument*. The decisive design choice that makes what it measures trustworthy is that we apply cumulative prospect theory (CPT) as a *reward transform after reward blending*, never inside the learning update (2): the optimiser is byte-for-byte identical across all conditions, so a measured shift in w_{crit} is a behavioral effect, not an artifact of tuning or of a changed algorithm—separating us cleanly from prior CPT-in-MARL work that folds the value and probability-weighting functions into the actor–critic update (Lalmohammed et al., 2025). Against this null, a three-way design (no PT / curvature only / curvature plus loss aversion) attributes any shift in w_{crit} to a specific behavioral component.

What we find is this. *Human-like loss aversion forces cooperative learning agents to require more reward centralization to avoid collapse—a shift we predict in sign a priori and confirm across three environments and two algorithms—and a single downstream loss-averse agent can set that requirement for the whole system.* The shift is already significant at the human-range $\lambda \approx 3$ ($+0.089$ $[0.041, 0.141]$ in required centralization), rising to a high-stakes upper bound of $+0.166$ $[0.135, 0.202]$ at λ=7; its *sign* is fixed before any data by a sign invariant of the policy-gradient escape barrier (Theorem 1); and its *direction* replicates across a supply chain (under PPO and centralized-critic MAPPO), MPE Cooperative Navigation, and Level-Based Foraging. Our contributions form a ladder: (i) the first quantitative, confidence-intervalled phase boundary of coordination collapse in deep cooperative MARL; (ii) the discovery that this boundary is *metastable*, sliding with the training horizon rather than being a fixed constant; (iii) a behavioral decomposition showing the two CPT components move it in *opposite* directions, with the loss-aversion direction generalizing and a *minority by network position* setting the boundary; (iv) an a priori sign guarantee for that direction; and (v) a practical reading of the boundary as a measurable safety margin—how much more centralization a cooperative system needs once its agents are loss-averse, and which participant sets the requirement. Two of these results (the affine λ-law and the downstream-minority cascade) we establish on the supply-chain substrate and scope honestly as such, framing the cascade as a falsifiable, structurally specified prediction for any directed-flow domain.

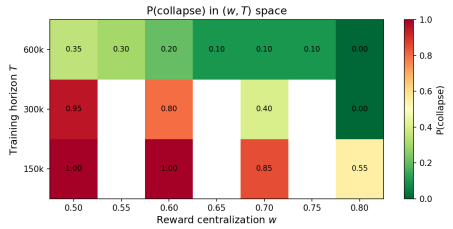


Figure 1: Collapse probability $P(\text{collapse})$ across the (w, T) plane; each cell is the empirical collapse fraction over 20 seeds. As the training horizon T grows, the collapse region recedes to lower w—the *metastable* leftward slide of the boundary. The 150k and 300k rows are risk-neutral (Exp A, α=β=1); the 600k row (Exp T, α=β=0.88) is partially curvature-inflated, and a matched α=β=1 control (Exp 0b) confirms the slide is genuine (§).

Preliminaries and Method

In cooperative MARL, N agents share an environment and aim to maximize a joint objective. Centralised Training with Decentralised Execution (CTDE) (Lowe et al., 2017) addresses the *credit assignment* problem by allowing a centralised critic during training while agents execute independently. The degree of credit sharing is controlled by a reward mixing weight: a scalar $w \in [0, 1]$ that interpolates between purely local (w=0) and purely global (w=1) reward.

When $w \approx 0$, each agent's reward is dominated by its own performance. The contribution of cooperative action is diluted: an agent receives no signal when a teammate's action improves the global outcome. Liu et al. (2023) and Shi and Peng (2023) formalise this as the *lazy agent* problem and *relative overgeneralization* respectively: agents converge to a safe, zero-action policy that avoids coordination risk. This paper asks: at what value of w does this failure mode become probable, and how does agent-level loss aversion shift that value?

Prospect theory (Kahneman and Tversky, 1979) models human decision-making under uncertainty via a reference-dependent value function and a nonlinear probability weighting function. Cumulative prospect theory (CPT) (Tversky and Kahneman, 1992) extends this to compound lotteries via rank-dependent integration. The CPT value function is:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0, \end{cases} \tag{1}$$

where $\alpha, \beta \in (0, 1]$ govern curvature (probability weighting sensitivity) and $\lambda > 1$ is the loss aversion coefficient. Setting α=β=1 and λ=1 recovers linear expected utility. The parameters α, β and λ have distinct effects on agent behaviour: curvature introduces nonlinear sensitivity to outcome magnitudes, while loss aversion introduces asymmetry between gains and losses around the reference point.

We apply CPT as a *reward transform* applied after reward blending. The per-step reward for agent i is:

$$r_i^{\text{PT}} = v\big((1 - w)\, r_i^{\text{local}} + w\, r^{\text{global}}\big)\,, \tag{2}$$

where $v(\cdot)$ is given by Equation (1). This is the decisive architectural difference from Lalmohammed et al. (2025), the only prior work bringing CPT into cooperative MARL: they fold *both* the value function and the probability-weighting function into the actor–critic update, which entangles the curvature (α, β) and loss-aversion (λ) effects and confounds any behavioral shift with a change to the optimiser. By transforming only the scalar reward and leaving the policy gradient untouched, our optimiser is *byte-for-byte identical* across all conditions: a measured shift in w_{crit} is therefore a behavioral effect, not an algorithmic one, and the curvature and loss-aversion channels can be dialed independently—which is what makes the component decomposition possible at all.

Measurement method

The reward centralization parameter $w \in [0, 1]$ is the sole structural intervention. All other environment parameters, algorithm hyperparameters, and random seeds are held constant across conditions.

We define coordination collapse as a binary outcome:

$$\text{collapsed}(\text{run}) = \mathbf{1}\left[\bar{S} < 10 \ \wedge \ \bar{\pi} \leq -127.1\right], \qquad (3)$$

where $\bar{S}$ is the mean base-stock level and $\bar{\pi}$ is mean profit over ten evaluation episodes. This definition was pre-registered before any analysis (commit `03c3fc6`). Section demonstrates robustness to $\pm 20\%$ threshold perturbation (maximum $\Delta w_{\text{crit}} = 0.015$).

For each experimental condition, we fit:

$$P(\text{collapsed} \mid w) = \sigma(b_0 + b_1 w), \qquad (4)$$

where σ is the logistic function. We estimate $w_{\text{crit}} = -b_0/b_1$ (the value of w where collapse probability equals 0.5) and its 95% confidence interval via 2,000 bootstrap resamples at the run level. with Wilson 95% intervals for per-w fractions. For steep transitions (high λ), where the unpenalized MLE slope is ill-conditioned, we use *Firth penalized* logistic regression (Jeffreys prior), stable in this regime; Firth and bootstrap w_{crit} agree throughout.

To attribute a shift in w_{crit} to a specific PT component, we compare three conditions with identical environment and algorithm: **null** ($\alpha{=}\beta{=}1, \lambda{=}1$), **curvature** ($\alpha{=}\beta{=}0.88, \lambda{=}1$: probability weighting, no loss asymmetry), and **loss aversion** ($\alpha{=}\beta{=}0.88, \lambda \in \{3, 7\}$: full PT). The curvature effect is $w_{\text{crit}}(\text{B}, \lambda{=}1) - w_{\text{crit}}(\text{N})$ and the loss-aversion increment $w_{\text{crit}}(\text{B}, \lambda) - w_{\text{crit}}(\text{B}, \lambda{=}1)$; the decomposition is valid because the null removes all PT distortion while curvature adds nonlinearity without loss asymmetry.

Experimental Setup

We use `NetworkManagement-v1` from or-gym (Hubbs et al., 2020): a six-node multi-echelon inventory network with Poisson demand ($\mu{=}20$) on the retail edge, 30-period episodes, and uniform lead times. Node 1 (the retail-facing node) follows a fixed demand-pull policy, stabilising the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterised via a base-stock wrapper that maps each agent's continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \text{IP})$ where IP is inventory position. The substrate extends prior DRL inventory work — the Beer Distribution Game DecPOMDP of Oroojlooyjadid et al. (2022), the multi-echelon A3C results of Gijsbrechts et al. (2022), and single-echelon MARL (Leluc et al., 2023) — to a six-node network with per-node agents.

Each agent runs PPO (Schulman et al., 2017) with a shared MLP policy (`n_steps=128`, `batch_size=256`) via Stable-Baselines3 (Raffin et al., 2021), PettingZoo (Terry et al., 2021), and SuperSuit (Terry et al., 2020) vectorisation. Evaluation is deterministic over 10 episodes with 20 seeds per condition.

Table 1 summarises all seven experiments. All designs were pre-registered before execution. Total run count: 1,524.

Table 1: Experiment summary (same environment and algorithm throughout; $\alpha{=}\beta{=}1$ is the linear no-PT condition). Total: 1,524 runs.

Exp	Purpose	w grid	λ	α, β	Runs
A@150k	Struct. boundary	0:0.1:1	1	1	220
A@300k	Struct. boundary	0.2:0.8	1	1	140
C	Basin control	0.5, 0.85, 1	1	1	144
N	Null baseline	0:0.2:1	1	1	180
B-Red.	Behav. decomp.	0.5−0.95	1, 3, 7	0.88	300
λ-sweep	Loss-av. law	0.5−0.95	2, 4, 5, 6	0.88	400
T	Metastab. horizon	0.5:0.8	1	0.88	140

Results

Structural Phase Transition and Metastability

Exp A establishes a sharp logistic collapse transition. At 150k steps, $w_{\text{crit}} = 0.822$ $[0.785, 0.858]$ with transition width $\Delta w = 0.278$ (logistic slope $b_1 = -15.78$, $p = 5 \times 10^{-9}$). At 300k steps, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ with $\Delta w = 0.199$ ($b_1 = -22.05$, $p = 3.2 \times 10^{-7}$). In both cases, collapse fractions rise steeply from near-zero to near-one within a $\Delta w < 0.3$ range, consistent with a genuine phase transition rather than a gradual degradation. We use "phase transition" throughout in the empirical sense of a sharp, metastable threshold—an exploration-driven, Kramers-style escape barrier (Section)—and not as a claim of a thermodynamic-limit singularity.

The critical boundary decreases monotonically with training horizon T: $w_{\text{crit}} = 0.822$ at 150k and $w_{\text{crit}} = 0.666$ at 300k (both from Exp A, $\alpha{=}\beta{=}1$). Figure 1 visualises the collapse surface in (w, T) space. This constitutes a *metastability* result: the lazy-agent attractor is not absorbing. The same leftward slide appears in *both* cross-environment substrates under centralized-critic MAPPO, so metastability is not a supply-chain artifact: on MPE w_{crit} falls $0.82 \to 0.68$ as T grows 2M $\to$ 8M (slide $+0.14$), and on Level-Based Foraging $0.33 \to 0.20$ as T grows 1M $\to$ 4M (slide $+0.13$), both with non-overlapping bootstrap intervals (Section). Agents trapped below the isocline at finite T can escape given sufficient training; w controls the energy barrier height of the attractor. The phase boundary proper is the asymptotic ($T \to \infty$) threshold—the deterministic crossing toward which

$w_{\text{crit}}(T)$ descends (Equation (9))—so the horizon dependence we observe is the metastable *approach* to that boundary, not evidence against a sharp underlying one. The sharpening transition width (Δw decreasing from 0.278 to 0.199 across 150k to 300k) is consistent with policy convergence reducing outcome variance.

The slide holds under matched $\alpha{=}\beta{=}1$: a control at 600k (Exp 0b) gives $P(\text{collapse})$ of 0.35/0.20 at $w{=}0.6/0.7$ versus 0.80/0.40 at 300k, confirming the horizon effect is real with curvature held fixed (the Figure 1 600k row mixes in the curvature shift; see caption). Initialization is ruled out as a driver: within-w correlations between S_{init} and converged S are near zero (-0.11 to -0.38; Exp C).

The structural null and its reproducibility

All behavioral shifts are measured against a *single canonical null*: the risk-neutral structural boundary at the 300k decomposition horizon, $w_{\text{crit}}^{\text{null}} = 0.666$ (CI $[0.635, 0.694]$, Exp A @300k), which the dedicated null experiment (Exp N) independently reproduces at 0.685 (difference 0.019, overlapping intervals). Every other null-like number is a labeled offset from this one value, not a separate baseline (Table 6): the 0.822 at 150k is the same null at a shorter, metastable horizon, and the 0.586 at $\alpha{=}\beta{=}0.88$ is a *result*—the curvature shift—and the $\lambda{=}1$ anchor of the loss-aversion sweep.

Behavioral Phase Boundary

We sweep loss aversion across the full range $\lambda \in \{1, 2, 3, 4, 5, 6, 7\}$ (all at $\alpha{=}\beta{=}0.88$, 300k steps, 20 seeds per cell), combining Exp B-Reduced at $\lambda \in \{1, 3, 7\}$ with a dedicated fill-in sweep at $\lambda \in \{2, 4, 5, 6\}$. The critical centralization w_{crit} rises with loss aversion across the entire range (Table 5, Figure 6): $w_{\text{crit}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$ for $\lambda = 1, \ldots, 7$. The increase is monotone up to estimation noise at the high-λ plateau: w_{crit} rises strictly from $\lambda{=}1$ to $\lambda{=}5$, then saturates near 0.84–0.85, where the $\lambda{=}5$ and $\lambda{=}6$ estimates (0.839 vs 0.836) are statistically indistinguishable (overlapping CIs).[1] The behavioral claim is unambiguous at the endpoints: $w_{\text{crit}}(\lambda{=}7)$ lower CI 0.845 exceeds $w_{\text{crit}}(\lambda{=}1)$ upper CI 0.624—non-overlapping by 0.221—and the seven-point trend (Figure 7) shows the relationship is systematic, not a two-point artefact.

Crucially, the high-λ plateau is a *behavioral* saturation, not an artifact of the boundary hitting the $w{=}1$ ceiling: the plateau sits at $w_{\text{crit}} \approx 0.85$ with clear headroom to $w{=}1$, and at $\lambda{=}7$ coordination is *fully* restored well before the ceiling (collapse fraction 0.00 already at $w{=}0.95$). A ceiling artifact would instead pile w_{crit} up against $w{=}1$ with collapse persisting at the highest w; we observe neither. The saturation therefore reflects diminishing marginal sensitivity of the coordination barrier to loss aversion—consistent with the bounded affine law (11), in which $w_{\text{crit}}(\lambda) \to 1$ only as $\lambda \to \infty$—rather than a measurement limit.

The per-w collapse curves for the full sweep appear in App. (Figure 6).

Table 2 and Figure 6 present the three-way decomposition. Relative to the null ($w_{\text{crit}} = 0.685$):

1. **Curvature effect** ($\alpha{=}\beta{=}0.88$, $\lambda{=}1$): excess $= -0.100$, 95% CI $[-0.172, -0.033]$. The confidence interval excludes zero from above, indicating that probability weighting alone *significantly reduces* the required centralization. Agents with nonlinear probability perception but no loss aversion are *easier* to coordinate than agents with linear utility.
2. **Loss aversion at** $\lambda{=}3$: excess $= +0.089$, 95% CI $[+0.041, +0.141]$ (CI excludes zero; significant).
3. **Loss aversion at** $\lambda{=}7$: excess $= +0.166$, 95% CI $[+0.135, +0.202]$ (CI excludes zero; significant).

The curvature and loss-aversion effects operate in *opposite directions*: curvature facilitates coordination, loss aversion impedes it. At $\lambda{=}7$, the loss aversion increment above the curvature baseline is $+0.265$ ($= 0.851 - 0.586$), overwhelming the curvature benefit of -0.100 for a net shift of $+0.166$ above the structural null (rounding: $-0.100 + 0.265 \approx 0.165 \approx 0.166$).

A Mean-Field Mechanism for the Collapse Boundary

This section explains *why* the boundary exists and how it moves. We claim it at the *sign-and-form level only*: w_{crit} marks a signal-to-noise crossing of the policy gradient, the two prospect-theory components move it in *opposite* directions, and the model predicts these directions, the monotonic ordering, the transition sharpness, and the sign of the horizon dependence *without fitting*—but not the numerical value of w_{crit}, which is calibrated (Table 7). The account is the learning-dynamics analogue of the EGT result of Mengesha et al. (2025): the driver is the gradient signal-to-noise ratio under exploration, not selection over fixed strategies, which makes it specific to gradient-trained neural policies.

Gradient flow and its noise

Each learning agent i runs PPO on a Gaussian policy over its (normalised) base-stock action $a_i \in [0, 1]$ with mean μ_i and exploration scale σ. The shaped reward is the centralisation blend passed through the prospect-theory value function v,

$$r_i = v\big((1-w)\,\ell_i(a_i, a_{-i}) + w\,G(a)\big), \qquad G(a) = \textstyle\sum_j \ell_j, \tag{5}$$

where ℓ_i is node i's local profit and G the system profit. For a Gaussian policy the PPO mean update is a stochastic estimate of the return gradient, so the policy mean follows a mean-field Langevin equation $\mathrm{d}\mu_i = D(w)\,\mathrm{d}t + \sqrt{2\,\Theta(w)}\,\mathrm{d}W$ with drift $D(w) = \eta\,\partial_{\mu_i}\mathbb{E}[(1{-}w)\ell_i + wG]$. Because each

[1] At high loss aversion the transition is very steep and falls between two adjacent grid points—e.g. collapse fractions $1.00, 1.00, 1.00, 0.55, 0.00$ at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$ for $\lambda{=}7$—so the boundary remains identifiable (the $w{=}0.85$ cell is interior, $11/20$ collapsed) but the *unpenalized* MLE slope diverges. We therefore fit every λ with Firth penalized logistic regression (Table 5), which is stable under steep transitions; all seven slopes are significant ($p < 10^{-3}$) and the Firth w_{crit} matches the bootstrap estimate to within 0.019. A finer grid ($w \in \{0.80, 0.90\}$) would tighten the high-λ CIs further.

Table 2: Three-way behavioral decomposition of w_{crit} relative to the null model. "Significant" denotes that the conservative difference confidence interval excludes zero. †For λ=7, w_{crit} is from bootstrap only; see footnote in Section .

Condition	w_{crit}	95% CI	Excess	Excess CI	Sig.?
Null (α=β=1, λ=1)	0.685	[0.657, 0.710]	0	—	—
α=β=0.88, λ=1	0.586	[0.539, 0.624]	-0.100	$[-0.172, -0.033]$	✓ (neg.)
α=β=0.88, λ=3	0.774	[0.751, 0.798]	$+0.089$	$[+0.041, +0.141]$	✓
α=β=0.88, λ=7†	0.851	[0.845, 0.858]	$+0.166$	$[+0.135, +0.202]$	✓

partner's local profit is blended into the global term, partner exploration enters agent i's gradient through the global channel, giving a centralisation-weighted, cross-agent diffusion

$$\Theta(w) \;\propto\; \tfrac{\sigma^2}{B}\big[(1-w)^2 s_\ell^2 \;+\; w^2(N-1)\,\kappa^2\big], \qquad (6)$$

with batch size B, local sensitivity s_ℓ, cross-agent sensitivity κ.

The boundary as a risk-dominance crossing

We instrumented the substrate directly (finite-difference of ℓ_i and G along the base-stock axis), and the asymmetry it reveals gives the drift a concrete game-theoretic reading. The continuous action induces a stochastic coordination game whose payoff topology shifts with w. At low w the *lazy* policy is the dominant strategy: an upstream node's inventory is a *local cost* (holding and procurement reduce its own profit). As w grows, a cooperative equilibrium emerges because that same inventory is a *global benefit* (it averts downstream stockouts), turning the environment into an (analogue of a) Stag Hunt. But the cooperative equilibrium emerging does not mean agents will *learn* it: if agent i explores cooperatively (raising a_i) while its partners stay lazy, it pays the local cost yet collects no global flow benefit—a continuous analogue of the *sucker's payoff*. Writing the linearised marginal incentives as $\partial_a \mathbb{E}\ell \to -c$ (the expected sucker penalty) and $\partial_a \mathbb{E}G \to +g$ ($c, g > 0$), the drift toward cooperation is

$$D(w) = \eta\big[\, w\, g - (1-w)\, c\,\big]. \qquad (7)$$

What governs whether the optimiser actually climbs toward cooperation is not the drift alone but its ratio to the gradient noise: escape from the lazy basin requires the cooperative equilibrium to become *effectively risk-dominant*—its basin of attraction wide enough for exploration noise to bridge—

$$\mathrm{SNR}(w) \;=\; \frac{D(w)}{\sqrt{\Theta(w)}} \;\gtrsim\; \kappa_c, \qquad (8)$$

for an $\mathcal{O}(1)$ critical value κ_c set by the training budget. The cooperative signal grows only with centralisation ($\propto w$), so for small w the right-hand region of (7) is swamped by the exploration variance (6) and the lazy policy is the effective attractor (Figure 2a). The crossing $\mathrm{SNR}(w) = \kappa_c$ defines w_{crit}. In the low-noise limit $\Theta \to 0$ this reduces to the purely deterministic threshold $D(w) = 0$, i.e.

$$w_{\text{crit}}^0 = \frac{c}{c+g}, \qquad (9)$$

the asymptotic ($T \to \infty$) low-noise *shadow* of the boundary; finite exploration noise raises the operative boundary above it. The drift zero $D(w)$=0 is not directly observable—a preregistered probe finds the conditioned drift weakly positive with no clean zero-crossing, an artifact of basin geometry (Section)—so we read the boundary as a *noise-bridged* risk-dominance crossing, not a drift-sign flip. The coefficients c, g, σ, κ_c are not derived from the cost matrix; the model is calibrated, and its content is the *functional* predictions below.

Sharpness and metastability

Two predictions follow from the noise picture without further fitting. First, escape is a Kramers first-passage problem with mean time $\tau \sim \tau_0 \exp(\Delta U/\Theta)$, exponentially sensitive to SNR; the collapse probability therefore switches abruptly as w crosses w_{crit}—an abrupt, metastable threshold, matching the steep logistic fits of Section . Second, collapse at finite horizon means $\tau > T$; as T grows, a smaller SNR (hence smaller w) suffices to escape, so $w_{\text{crit}}(T)$ decreases monotonically toward (9). This is exactly the observed metastable slide $w_{\text{crit}} = 0.822 \to 0.666$ from 150k to 300k steps, following the $\propto -\ln T$ Kramers trend (Figure 2c).

Formal account: what is predicted versus calibrated

Let $x \in [0,1]$ be a macroscopic *order parameter*—the fraction of agents in the coordinating configuration—evolving under the policy-gradient flow as the Itô SDE

$$\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, N)}\,\mathrm{d}W_t, \qquad (10)$$

where F is the mean-field drift of (7) (so $\partial_w F > 0$ and $\partial_\lambda F < 0$ follow from the payoff structure, not assumption: w weights the cooperation gradient $+g$ while λ scales only the loss domain) and $\Sigma(x, N) > 0$ is the gradient-noise variance. Here x is *latent*—not recoverable from policy statistics (Section)—so the claims below concern its law, not a measured trajectory. Three results delimit what the mechanism predicts a priori. Two of them (Theorem 2, mean-field criticality as $N \to \infty$; Theorem 3, the finite-(N,T) calibration gap) establish that the *magnitude* of w_{crit} is calibrated, not predicted; we state and prove them in App. . The third—the one the central claim rests on—we give here.

Theorem 1 (Sign invariance of the loss-aversion shift)**.** *Assume* $\Sigma(x,N) > 0$ *and* $\partial_\lambda F(x,w,\lambda) < 0$ *for all* $x \in [x^*_{\text{saddle}}, x^*_{\text{coord}}]$. *Then the empirical boundary shifts strictly in the direction the mean-field drift predicts:* $\partial_\lambda w_{\text{crit}}^{\text{Emp}} > 0$.

Proof. Write (12) as a forward integral, $\Delta U = \int_{x^*_{\text{saddle}}}^{x^*_{\text{coord}}} 2F(z,w,\lambda)/\Sigma(z,N)\,\mathrm{d}z$ (the two minus signs in (12)—the integrand sign and the reversed limits—cancel).

Differentiating under the integral, the fixed-point locations contribute nothing to first order ($F = 0$ there), so

$$\partial_\lambda \Delta U = \int_{x^*_{\text{saddle}}}^{x^*_{\text{coord}}} \frac{2\,\partial_\lambda F(z, w, \lambda)}{\Sigma(z, N)}\,\mathrm{d}z < 0,$$

the inequality holding *pointwise* because $\Sigma > 0$ and $\partial_\lambda F < 0$ on the whole interval—independently of the sign of F itself. A lower barrier shortens the escape time, $\partial_\lambda \tau < 0$; symmetrically $\partial_w F > 0$ raises the barrier, $\partial_w \tau > 0$. Applying the implicit function theorem to $G(w, \lambda) = \tau(w, \lambda) - T = 0$,

$$\frac{\mathrm{d}\, w^{\text{Emp}}_{\text{crit}}}{\mathrm{d}\lambda} = -\frac{\partial_\lambda \tau}{\partial_w \tau} = -\frac{(-)}{(+)} > 0.$$

Thus the *magnitude* depends on Σ (Theorem 3), but the *direction* is guaranteed by the mean-field drift alone—the precise sense in which the mechanism is predictive. The identical argument with ∂F signs reversed for concave–convex curvature gives the opposite shift, yielding the predicted directional divergence of the two prospect-theory components. □

Opposite signs of the two prospect-theory components

The mechanism's most distinctive prediction is qualitative and requires no fitting: probability weighting and loss aversion shift w_{crit} in *opposite* directions, and the risk-dominance reading makes the asymmetry concrete. Concave–convex curvature ($\alpha, \beta < 1$) compresses the magnitude of both gains and losses, smoothing the extremes of the stochastic sucker's payoff; this lowers the outcome variance that enters Θ, *raises* the SNR, and hence *lowers* w_{crit}. Loss aversion ($\lambda > 1$) instead magnifies the downside of the sucker's payoff, deepening the perceived lazy basin, inflating Θ and lowering the SNR, which *raises* w_{crit}. The signs are fixed by the value function before any data are seen, and they match the central empirical finding (Section): curvature alone shifts w_{crit} left by 0.100, loss aversion shifts it right by up to 0.166.

Derived form of the loss-aversion law

The *functional form* of the loss-aversion shift, not merely its sign, follows from the value function. Differentiating $\mathbb{E}[v(r)]$ and splitting the marginal mass into gain and loss domains, the loss domain is the only part scaled by λ, so the effective local term is affine in λ:

$$w_{\text{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \qquad \phi(\lambda) = 1 + \rho\,(\lambda - 1), \quad (11)$$

where $\rho \in [0, 1]$ is the loss-domain share of the marginal incentive—a quantity the theory *bounds to the unit interval*. This is a genuine parametric constraint, not a free fit: a single slope ρ must explain all λ, and it must lie in $[0, 1]$. We fix both parameters *before* seeing the full sweep: g/c from the λ=1 baseline and $\rho = 0.61$ from the three Exp B-Reduced points ($\lambda \in \{1, 3, 7\}$), both of which satisfy the bound. Held fixed and tested *out of sample* against the remaining four points ($\lambda \in \{2, 4, 5, 6\}$; the full seven-point sweep in Table 5), this single-slope affine law gives

$$w^{\text{pred}}_{\text{crit}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\},$$
$$w^{\text{meas}}_{\text{crit}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\},$$

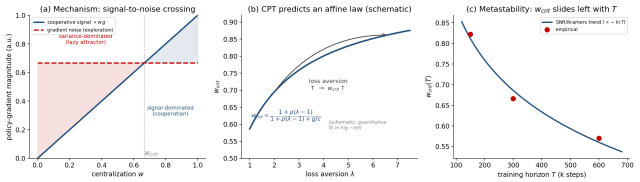


Figure 2: Mean-field mechanism versus empirics. (a) Mechanism (schematic): the cooperative gradient signal grows with centralisation ($\propto w\,g$) while partner exploration injects a roughly w-independent noise floor; w_{crit} is the signal-to-noise crossing (8), below which the lazy policy is the effective attractor. (b) Schematic of the *derived affine* loss-aversion law (11): CPT predicts $\phi(\lambda) = 1 + \rho(\lambda - 1)$ with slope $\rho \in [0, 1]$; the single data-implied slope $\rho = 0.61$ respects the bound. The quantitative fit against the full seven-point sweep is shown separately in Figure 7. (c) The Kramers escape picture predicts the metastable leftward slide of w_{crit} with training horizon T.

an RMSE of 0.014 across all seven points (Figure 7)—a one-parameter, theory-bounded law tracking the entire $w_{\text{crit}}(\lambda)$ curve, not merely the three calibration points (re-fitting ρ to all seven gives $\rho = 0.63$, RMSE 0.013). The lone residual is a mild overshoot at the λ=6, 7 plateau: the leading-order law is affine while the data saturate mildly, a second-order effect we do not model. We report this derived affine law rather than a better-fitting but unmotivated power law because the point is mechanism, not goodness of fit.

Scope. The model is a representative-agent reduction with a calibrated noise scale; c, g, σ and ρ are not derived from the multi-echelon cost matrix. A direct finite-difference probe of the substrate confirms the *sign* of the local-cost/global-benefit asymmetry but shows the marginal coefficients are operating-point dependent, so the absolute boundary is a property of the stochastic multi-agent dynamics rather than a static gradient—which is why we calibrate rather than predict its magnitude. This section is not the paper's contribution— the decomposition is—but a supporting, falsifiable *mechanism*: it commits in advance to the signs, the ordering, the affine λ-law with a bounded slope, the sharpness, and the horizon and exploration-noise directions (Table 7), all of which the data bear out, consistent with exploration-driven accounts of cooperation failure in shared-policy MARL (Weng and Lee, 2026). The exploration-noise prediction is confirmed directly by the entropy ablation (Section). Section reports a direct test that measures the Langevin drift and diffusion from PPO training: it reproduces the matched-horizon boundary but leaves the magnitude calibrated, as the drift-sign observable is confounded by basin geometry. We therefore claim this mechanism at the *sign-and-form level only*—the opposite signs, the monotone ordering, the affine form, and the horizon and exploration-noise directions—and treat the numerical value of w_{crit} as calibrated throughout. No result in this paper rests on a predicted magnitude; upgrading the account from calibrated to predictive is left to future work.

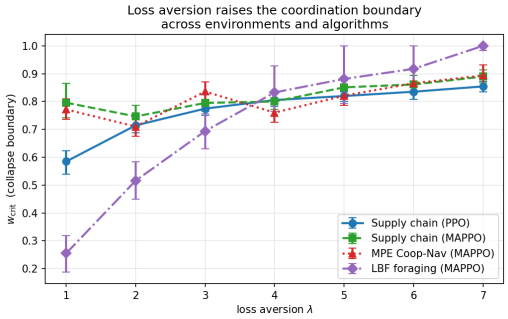


Figure 3: **The loss-aversion effect generalizes.** Critical centralization w_{crit} versus loss aversion λ for four settings: the supply chain under PPO (blue) and under centralized-critic MAPPO (green), MPE Cooperative Navigation under MAPPO (red), and Level-Based Foraging under MAPPO (purple). Error bars are bootstrap 95% CIs. Across three structurally distinct environments and two algorithms, w_{crit} rises with λ then saturates—loss aversion raises the centralization a cooperative system needs to avert collapse. The *direction* is what generalizes; absolute levels differ (most at λ=1) and the curvature sign and exact functional form are substrate-specific. (LBF's high-λ points saturate against the centralization ceiling and are reported as exploratory; see Section .)

Generalization across Environments and Algorithms

The mechanism of Section commits, before any data, to a *direction*: loss aversion inflates gradient noise and therefore raises w_{crit}. If that account is right rather than an artefact of one substrate or one optimiser, the sign should survive changing both. We test this on two further environments and a second algorithm. The headline result is that the *direction* of the loss-aversion effect is robust to both, while the absolute boundary, the curvature sign, and the precise functional form are substrate- and algorithm-specific—exactly the split the calibrated-not-predicted mechanism (Table 7) anticipates. Table 8 (Appendix) records the breadth of evidence behind every headline claim. Figure 3 summarises the headline: in all four settings w_{crit} climbs with λ, the curves converging in the mid-range and differing most at λ=1.

Across environments: incentive misalignment across domains

We test whether the boundary—and the loss-aversion direction—is a recurring feature of incentive-misaligned multi-agent systems rather than a supply-chain artefact, in two structurally distinct environments. Both share the reward blend $r_{\text{blend}}(i) = (1-w)\, r_{\text{local}}(i) + w\, r_{\text{global}}$ but reach collapse by different routes: the supply chain by *resource competition* (hoarding starves downstream nodes) and MPE Cooperative Navigation (`simple_spread_v3`) by *coordination failure* (three agents herd together and cover no landmark). For MPE, collapse is measured by per-step landmark occupancy—a behavioural metric invariant to reward scale, hence unaffected by the CPT transform. Because MPE's coordination reward is *shared-global*, independent learners confound behavioural collapse with a credit-assignment failure; we therefore train it with centralized-critic MAPPO (Yu et al., 2022) (actor sees local observation plus a one-hot agent ID, critic sees the joint state—textbook CTDE), matching the algorithm to the environment's learnability so the measured boundary reflects the incentive w, not the learner's capacity to assign credit. We set the herding scale so the null boundary lands interior (w_{crit}=0.74) and run the full sweep $\lambda \in \{1, \ldots, 7\}$ at α=β=0.88 (1,760 runs; Figure 3, per-w curves in App.).

The *loss-aversion* effect replicates. On MPE (full λ=1…7 sweep, 20 seeds per cell), w_{crit} rises with loss aversion from 0.78 (λ=1) to 0.89 (λ=7)—rising in envelope but non-monotone at intermediate λ (the λ=2, 4 points dip to within seed noise of the null), so the effect is read as a direction, not a strict ordering. The loss-aversion excess over the null is $+0.158$ $[+0.108, +0.212]$ at λ=7 and $+0.101$ $[+0.055, +0.150]$ at λ=3, both significant and the latter quantitatively matching the supply-chain $+0.089$. The qualitative law—loss aversion raises the centralization required to avert collapse—thus holds across structurally distinct environments and training algorithms (a third environment, Level-Based Foraging, is added in Section ; Figure 3).

The *curvature* effect does *not* replicate. On MPE the curvature excess is $+0.042$ $[-0.009, +0.092]$, statistically indistinguishable from the null and opposite in sign to the supply-chain -0.100; probability weighting appears to facilitate coordination only in the supply-chain substrate. The exact *functional form* of the loss-aversion law is likewise environment-specific: the affine $w_{\text{crit}}(\lambda)$ that tracks the supply chain to RMSE 0.014 fits MPE only to RMSE 0.05, the MPE curve being steeper at high λ as the boundary approaches the centralization ceiling. We therefore claim cross-environment generality for the collapse boundary and for the *direction* of the loss-aversion effect—the paper's central behavioural claim—while reporting the curvature sign and the precise functional form as supply-chain-specific.

A third environment: Level-Based Foraging

MPE shares MAPPO's shared-global reward; to probe a third, structurally distinct domain we add Level-Based Foraging (LBF), a standard cooperative-MARL benchmark (Christianos et al., 2020) with a *public-goods* incentive: agents pay an effort cost to load food cooperatively, so the locally rational action is to free-ride and the team coordinates only as the global reward comes to dominate. LBF is natively discrete and unrelated to logistics or navigation; we train it under centralized-critic MAPPO—matching MPE, so that across MPE and LBF the *algorithm* is held fixed and only the *environment* varies—with the identical reward pipeline and full λ-sweep, and a collapse threshold set by the same collapsed-vs-coordinated midpoint used for MPE (1,600 runs, $w \in \{0, \ldots, 0.95\}$ including the w=0.90 resolver).

LBF yields the most complete decomposition outside the supply chain (Figure 3). The risk-neutral null has a low boundary ($w_{\text{crit}} \approx 0.08$); curvature moves it *right* by $+0.18$—*opposite* to the supply chain's -0.10, again showing the curvature sign is substrate-specific. Loss aversion then drives

a large rightward shift: w_{crit} rises from 0.26 (λ=1) to 0.83 (λ=4), monotone with non-overlapping low- versus high-λ intervals (the pre-registered criterion). For $\lambda \geq 5$ the boundary saturates against the centralization ceiling and we report those points as exploratory (a w=0.90 resolver sharpens the mid-λ estimates but does not change this). As in MPE, the *direction* replicates while curvature sign and functional form do not: the affine law misfits LBF by RMSE 0.15 (versus 0.014 on the supply chain), because the LBF boundary rises more steeply in λ. The *method*—a CI'd boundary that loss aversion shifts—generalizes; the affine *constants and form* are substrate-specific.

Across algorithms: MAPPO on the supply chain

MPE varies environment *and* algorithm at once. To isolate the algorithm, we re-run the full sweep on the *identical* supply-chain substrate, grid, seeds, and label, changing only the learner from independent PPO to centralized-critic MAPPO (Yu et al., 2022) (700 runs). The law replicates: w_{crit} rises from 0.796 [0.742, 0.860] at λ=1 to 0.889 [0.856, 0.917] at λ=7, monotone from λ=2 on, non-overlapping endpoints (Figure 3). As in MPE the *direction* generalizes while the absolute level does not (the two boundaries converge by $\lambda \geq 4$, $|\Delta w_{\text{crit}}| \leq 0.04$).

The mechanism's noise prediction: a dissociation

The mechanism makes a sharper second prediction: injected exploration noise should move the boundary *only where it is the binding term* in the SNR (8). This dissociates as predicted—the entropy effect replicates on the supply chain and LBF (where the landscape is sparse and injected noise binds) but is a *mechanism-predicted null* on MPE, whose dense reward and high intrinsic exploration swamp the added term (pooled slope β=+0.47, 95% CI [−3.4, +4.3], 1,320 runs). Metastability likewise replicates on both transfer environments. The effect appears exactly where the mechanism says it should and is absent where it says it should be—a stronger test than uniform replication (App.).

Two ablations confirm the boundary is contingent on the ingredients the mechanism names: it disappears under coarse discretisation of the continuous control (removing the policy gradient) and is measurable only with a credit-assignment-capable learner on shared-reward tasks (App.). Finally, the boundary is not an averaged property: a single *downstream* loss-averse node can set the centralization requirement of the whole network, a cascade that propagates upstream along the ordering chain (App. ; discussed below).

Discussion

That probability weighting and loss aversion exert opposite effects on the coordination boundary is, to our knowledge, novel; prior work applies prospect theory as a unified model without decomposing its components (Lalmohammed et al., 2025; Ghaemi et al., 2024). Our null-model design isolates them: on the supply chain the beneficial curvature effect (−0.100, substrate-specific) is masked when combined with loss aversion, whose centralization-raising effect dominates and is the component we establish as general. The operational reading is concrete: a system with λ=7 agents—bracketing human loss-aversion estimates (Tversky and Kahneman, 1992)—requires $w > 0.851$ for reliable coordination against $w > 0.685$ for risk-neutral agents, so the additional 0.166 (Table 2) is a quantified cost of deploying loss-averse agents.

Two caveats sharpen this reading. On calibration, the canonical laboratory estimate of loss aversion is $\lambda \approx 2.25$ (Tversky and Kahneman, 1992); our sweep deliberately brackets the empirical human range, with λ=7 an upper, high-stakes bound rather than a typical value. The cost scales with λ (Table 2), so a domain whose participants sit nearer $\lambda \approx$ 2–3 incurs a smaller but still positive margin (the λ=3 excess is +0.089, against +0.166 at λ=7). On deployment, centralization is not free: privacy, communication cost, and organisational policy can cap the attainable w. Where the required w exceeds that cap, the boundary reframes the design problem—instead of only raising centralization, one can lower the agents' effective loss aversion (for instance through the training objective), treating λ as the second, behavioural lever the decomposition exposes.

The boundary is also not an averaged property. App. shows it is set by a *minority*: a single downstream loss-averse node moves the system boundary as far as a fully loss-averse population, cascading upstream along the ordering chain. The operational consequence is sharp—in a mixed human–agent supply chain, the required centralization is governed by the most downstream loss-averse participant, not by the population mean.

The sliding w_{crit} is a positive finding: the lazy attractor is *metastable*, so agents below the isocline at finite T can coordinate with more training—consistent with Weng and Lee (2026)—and the minimum w therefore depends on the training budget.

Our framing sits within a recognised line of multiagent phase-transition results but differs in object: Yamaguchi (2025) chart a phase diagram for IQL over *environmental* axes, and Capannoli et al. (2026) prove a metastable boundary in opinion dynamics under a *fixed update rule*; ours is the deep-MARL learning-dynamics analogue, a boundary set by the policy-gradient signal-to-noise ratio that a behavioral bias then moves.

Several limitations bound these claims. *Substrate scope:* the full quantitative characterization (decomposition, affine λ-law, metastability, cascade) uses one substrate, chosen because its continuous, per-agent economic reward makes w_{crit} finely identifiable; we replicate the boundary and the loss-aversion *direction* in two further environments (MPE, LBF) and under a second algorithm on the same substrate, but the curvature sign and the exact functional form do *not* transfer, so we claim only the loss-aversion direction as cross-substrate. *Sweep scope:* the steep high-λ transitions are resolved by a single interior grid point (handled with Firth fits, Table 5); the metastability result is finite-horizon; and the heterogeneity sweep uses binary $\lambda \in \{1, 7\}$. *Method scope:* the approach needs an environment learnable without experience sharing and a collapse label separable from task-unlearnability, excluding sparse, long-credit-horizon tasks.

Cascade scope: the downstream cascade is single-substrate by structural necessity—it is well-posed only where a directed resource-flow ordering, genuine local-vs-global tension, and a learning-reachable boundary co-occur, a rare conjunction most benchmarks violate (App.); we therefore frame it as a falsifiable, structurally specified prediction rather than an established generality.

Conclusion

We presented the first quantitative phase diagram of cooperative MARL coordination collapse, characterising the critical centralization threshold w_{crit} as a function of reward structure, training horizon, and prospect-theoretic loss aversion. Our central and most strongly supported finding is that *loss aversion* impedes coordination—raising w_{crit} ($+0.089$ at $\lambda{=}3$, within the human range; up to $+0.166$ at $\lambda{=}7$)—a direction we predict mechanistically a priori and confirm across three environments and two algorithms. Scoped to the supply-chain substrate, two further results sharpen the picture: a three-way decomposition shows probability-weighting curvature acting in the *opposite* direction (easing coordination by 0.100), and the boundary is set by a downstream loss-averse *minority* rather than the mean bias. We present the latter as a striking single-substrate result and frame it as a falsifiable prediction for any directed-flow domain, rather than a claim shown to generalize. These results provide concrete design constraints for deploying cooperative MARL in behavioral settings and establish a pre-registered phase-diagram methodology applicable to any cooperative MARL system.

Natural extensions include replication across further algorithms (QMIX) and environments (SMAC), upgrading the mean-field account (Section) from a calibrated mechanism to a quantitative prediction of w_{crit} via direct measurement of the policy-gradient signal-to-noise ratio (Section), adaptive centralization scheduling based on the estimated isocline, and characterization of heterogeneous-λ populations.

Boundary Conditions: Two Ablations

The mean-field account (Section) makes the boundary contingent on two ingredients: a *continuous* policy gradient (the signal-to-noise crossing is a property of gradient flow), and a learner able to separate behavioural collapse from credit assignment (so that what w moves is the incentive, not the optimiser's capacity to coordinate). We ablate each. Both confirm the mechanism and delimit *where* the phenomenon lives.

Action representation

The mechanism predicts that destroying the continuous policy gradient should remove the boundary. Value-based learners require a discrete action space, so we discretised the supply-chain base-stock control into $K{=}11$ bins and ran an independent value learner (IQL) and a centralised monotonic-mixing learner (QMIX) on the identical reward pipeline, w-grid, and collapse label. Both produced a *flat, w-independent* collapse rate with no fittable w_{crit} in any condition. To isolate the cause—value-based learning vs. the discrete action space—we add a single-variable control: PPO, the learner that yields the boundary under continuous control, trained on the *same* $K{=}11$ discrete shell with everything else fixed. It too goes flat (Table 3). Holding the learner at PPO, discretisation *alone* erases the boundary, so the value-based nulls require no appeal to the learning paradigm.

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes (Section)
QMIX (centralised value)	discrete $K{=}11$	no
IQL (independent value)	discrete $K{=}11$	no
PPO (control)	discrete $K{=}11$	**no**

Table 3: Action-representation ablation. The control (last row) changes only the action space while holding the learner at PPO, isolating coarse discretisation of the continuous control—not the value-based paradigm—as what erases the boundary.

We scope this precisely. The supply-chain action is a *continuous* base-stock level; coarse bins snap the policy to a stable interior level and never visit the near-threshold (s, S) region where collapse occurs, exactly as the continuous-gradient mechanism predicts. This is *not* a claim that discreteness per se removes the boundary: MPE (Section) has a *natively* discrete action space and retains it. The effect is therefore a coarse-discretisation boundary condition, substrate-specific in the same way the curvature sign is—and a confirmed prediction of the mechanism, not a negative result.

Credit assignment

The second ingredient is a learner that can coordinate at all on a shared-reward task, so that w_{crit} reflects the behavioural incentive rather than a credit-assignment failure. We ablate the centralised critic. On MPE Cooperative Navigation, an independent learner (PPO) never reaches the coordinated regime at *any* w: the collapse fraction is 1.0 across the entire w-range, including maximal centralisation (Figure 4), so behavioural collapse cannot be distinguished from the inability to assign credit. A centralised critic (MAPPO) on the identical environment recovers coordination—the collapse fraction falls to 0.35 and landmark occupancy crosses the calibrated threshold near $w{\approx}0.6$. We observe the same independent-learner floor on Level-Based Foraging, where PPO fails to coordinate while MAPPO reaches the coordinated regime. The boundary is therefore *measurable* only with a centralised critic on shared-reward coordination tasks; this is why we match a centralised critic to such environments (Section)—a necessity, not a convenience.

A Minority Sets the Boundary (Single-Substrate Result)

The decomposition and its generalization treat the population as homogeneous—every agent carries the same λ. A deployed cooperative system rarely is: some participants are loss-averse, others are not. We therefore ask whether the boundary is an averaged, population-wide property, or whether a *minority* can set it—and whether *network position*

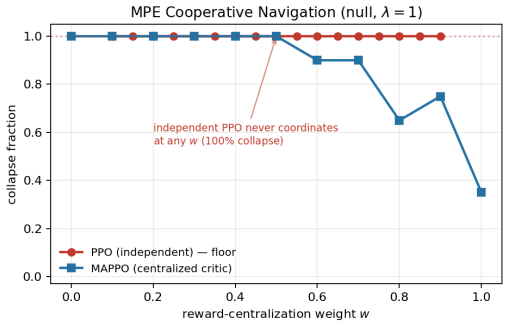


Figure 4: Credit-assignment ablation on MPE (null, λ=1). Independent PPO collapses at every w (a coordination floor: it cannot learn the task regardless of incentive), while the centralised-critic learner (MAPPO) crosses the boundary. The behavioural boundary is exposed only once credit assignment is handled.

matters. We assign λ=7 to selected nodes and λ=1 to the rest, holding the substrate, PPO configuration, collapse label, and grid identical to the main sweep (α=β=0.88, 300k, 20 seeds, $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$).

The effect is large and strongly position-dependent (Figure 5, Table 4). A single *downstream* loss-averse node drives $w_{\text{crit}} = 0.853$ $[0.849, 0.858]$—statistically indistinguishable from the *fully* loss-averse population ($w_{\text{crit}} = 0.851$ $[0.847, 0.856]$): one biased node downstream sets the centralization requirement of the entire network. The same node placed *upstream* shifts the boundary far less, to $w_{\text{crit}} = 0.726$ $[0.689, 0.763]$. Loss aversion therefore cascades *upstream*—a downstream node's bias propagates back along the ordering chain and dominates the system boundary—but not downstream, a directional asymmetry that mirrors the bullwhip structure of the substrate.

That a behavioral minority can move a system-level tipping point is itself established in evolutionary game theory via replicator selection (Mengesha et al., 2025); our contribution is the deep-MARL instantiation, where the effect emerges from gradient-learning dynamics and carries a concrete network-position signature absent from the EGT account. The operational reading is sharper than a population average would suggest: in a mixed human–agent supply chain, the centralization the system needs is set by its most downstream loss-averse participant, not by the mean bias.

Table 4: Heterogeneous-population sweep (α=β=0.88; loss-averse nodes at λ=7, others λ=1; 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs. Homogeneous references: all risk-neutral ≈ 0.586, all loss-averse 0.851.

Loss-averse nodes	w_{crit} (95% CI)
one upstream	0.726 [0.689, 0.763]
one downstream	0.853 [0.849, 0.858]
two downstream	0.851 [0.847, 0.856]
all five (homogeneous)	0.851 [0.847, 0.856]

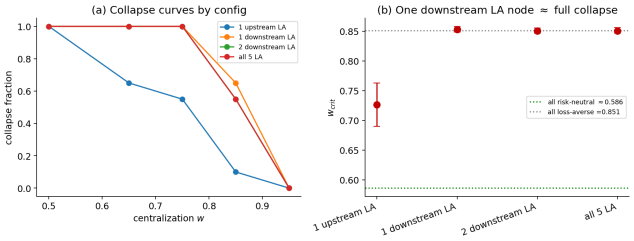


Figure 5: Heterogeneous cognitive profiles. (a) Collapse curves: a single downstream loss-averse node (orange) tracks the fully loss-averse population (red), while a single upstream node (blue) is far milder. (b) w_{crit} by configuration: one downstream loss-averse node already reaches the homogeneous loss-averse boundary; loss aversion cascades upstream but not downstream.

Table 5: Full loss-aversion sweep (α=β=0.88, 300k steps, 20 seeds per (w, λ) cell, n=100 runs per λ). w_{crit} with bootstrap 95% CIs. Slope and p are from *Firth penalized* logistic regression (Jeffreys-prior), which is stable under steep transitions; all seven slopes are significant ($p < 10^{-3}$). The Firth $w_{\text{crit}} = -b_0/b_1$ agrees with the bootstrap estimate at every λ (max. deviation 0.019), confirming that the very large unpenalized MLE slopes previously seen at λ=5, 7 were a fitting artifact, not separation.

λ	w_{crit}	95% CI	Firth slope b_1	p
1	0.586	[0.539, 0.624]	-14.7	1.2×10^{-6}
2	0.715	[0.687, 0.745]	-27.2	4.4×10^{-6}
3	0.774	[0.751, 0.798]	-29.7	5.1×10^{-6}
4	0.806	[0.766, 0.841]	-36.8	5.5×10^{-6}
5	0.839	[0.788, 0.845]	-53.1	4.7×10^{-4}
6	0.836	[0.806, 0.860]	-24.7	2.5×10^{-6}
7	0.851	[0.845, 0.858]	-43.1	1.5×10^{-3}

Supporting Tables

These reconciliation and claim-ledger tables support the main text.

Notation and Abbreviations

Abbreviations. MARL (multi-agent reinforcement learning); CTDE (centralized training, decentralized execution); CPT (cumulative prospect theory); PPO (proximal policy optimization); MAPPO (multi-agent PPO with a centralized critic); MPE (multi-agent particle environments); SNR (signal-to-noise ratio); MLE (maximum-likelihood estimate); CI (confidence interval).

Claim–Evidence Ledger

Table 8 states exactly how broadly each headline claim is supported, so the scope of every claim is legible at a glance. The loss-aversion direction is the generalizing core (three environments, two algorithms); the curvature sign and the downstream-minority cascade are supply-chain-specific, the

Table 6: The recurring w_{crit} values and the single role each plays. There is one structural null; the others are a horizon measurement of it and a curvature *result*, not independent baselines.

w_{crit}	Condition	Role
0.666	$\alpha{=}\beta{=}1$, 300k	canonical null
0.685	$\alpha{=}\beta{=}1$, 300k (Exp N)	null, reproduced
0.822	$\alpha{=}\beta{=}1$, 150k	null at shorter horizon
0.586	$\alpha{=}\beta{=}0.88$, $\lambda{=}1$	curvature result

Table 7: What the mean-field mechanism claims. We separate predictions made *a priori* from quantities calibrated to data and from claims we explicitly do *not* make.

Derived (no fit)	opposite signs of curvature vs. loss aversion; monotonic $w_{\text{crit}}(\lambda)$; *affine* form of $\phi(\lambda)$ with slope $\rho \in [0,1]$; abrupt (phase-transition) sharpness; w_{crit} decreasing with horizon T; w_{crit} increasing with exploration noise.
Empirically confirmed	the opposite signs and affine form (Section); the horizon slide (Exp A/T), which also generalizes to MPE and LBF; and the exploration-noise dependence—raising the entropy coefficient moves w_{crit} right (supply chain and LBF; a mechanism-predicted null on MPE, where intrinsic exploration dominates), collapsing the system at large noise (Section , Section).
Calibrated (1 param/axis)	g/c (from the $\lambda{=}1$ boundary); loss-domain share ρ; noise scale setting the absolute w_{crit}.
Not claimed	a closed-form value of w_{crit} from the cost matrix; the numerical magnitude on any axis; the mild concavity of $w_{\text{crit}}(\lambda)$ beyond leading order.

latter framed as a falsifiable prediction for directed-flow domains (Section).

Threshold Robustness

We verify that the collapse definition (Equation (3)) is robust to perturbation of its thresholds. We re-estimate w_{crit} on Exp A @300k data under nine variants of the definition, varying $\bar{S}$ threshold over $\{8, 10, 12\}$ and $\bar{\pi}$ threshold over $\{-102.5, -127.1, -152.5\}$. Three variants with threshold $\bar{\pi} \leq -152.5$ are degenerate (the minimum profit in the dataset is -128.1, so no run satisfies this condition); they are excluded. Across the remaining six estimable variants, the maximum deviation from the baseline $w_{\text{crit}} = 0.666$ is 0.015, well below the confidence interval width. We conclude that the collapse definition is robust to $\pm 20\%$ threshold perturbation.

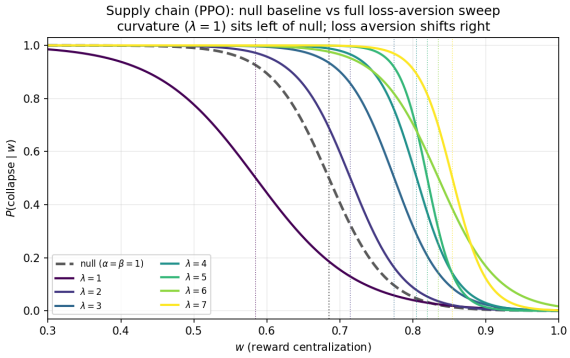


Figure 6: Behavioral decomposition and full loss-aversion sweep on the supply chain (PPO, 300k steps, 20 seeds per cell). Each curve is the Firth-fitted $P(\text{collapse} \mid w)$; vertical dotted lines mark w_{crit}. The null ($\alpha{=}\beta{=}1$, gray dashed) is the structural baseline ($w_{\text{crit}}{=}0.685$). Probability-weighting *curvature* alone ($\lambda{=}1$, $\alpha{=}\beta{=}0.88$) shifts w_{crit} *left* of the null—facilitating coordination—while *loss aversion* ($\lambda{=}2, \ldots, 7$, $\alpha{=}\beta{=}0.88$) shifts it *right* (rising, then saturating). The two prospect-theory components thus move the boundary in opposite directions, and the full sweep traces the resulting $w_{\text{crit}}(\lambda)$ progression in one panel.

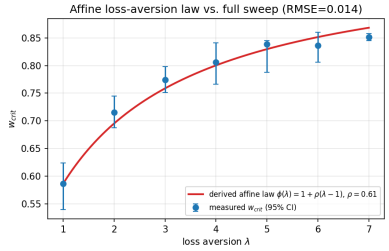


Figure 7: The derived affine loss-aversion law (11) against the full sweep. Points are the measured $w_{\text{crit}}(\lambda)$ for $\lambda = 1, \ldots, 7$ with bootstrap 95% CIs (Table 5); the line is the one-parameter law with g/c fixed at the $\lambda{=}1$ baseline and a single theory-bounded slope $\rho = 0.61$. The law tracks all seven points to RMSE 0.014, with a mild concave residual at the high-λ saturation plateau.

Symbol	Meaning
w	reward centralization weight, $\in [0, 1]$
w_{crit}	critical w where $P(\text{collapse})$=0.5
λ	loss-aversion coefficient (λ>1)
α, β	CPT value-function curvature (gain/loss)
ρ	loss-domain share of marginal incentive, $\in [0, 1]$
g, c	marginal global benefit / local cost of cooperation
$D(w), \Theta(w)$	Langevin drift / diffusion of the policy mean
$\text{SNR}(w)$	$D(w)/\sqrt{\Theta(w)}$, escape signal-to-noise ratio
σ, η, B	exploration scale / learning rate / batch size
$\Delta U, \tau$	escape barrier / Kramers first-passage time
T	training horizon (environment steps)
$\bar{S}, \bar{\pi}$	mean base-stock level / mean profit (collapse label)

Table 8: Each headline claim and the breadth of evidence behind it. Environments: SC (supply chain), MPE, LBF. Algorithms: PPO, MAPPO.

Claim	Envs	Algos
Sharp collapse boundary w_{crit} exists	SC, MPE, LBF	PPO, MAPPO
Metastability (w_{crit} slides with T)	SC	PPO
Loss aversion *raises* w_{crit} (dir.)	SC, MPE, LBF	PPO, MAPPO
Affine λ-law (functional form)	SC	PPO
Mechanism fixes sign a priori	SC, entropy abl.	PPO
Curvature *lowers* w_{crit} (sign)	SC (opp. on LBF)	PPO
Downstream-minority cascade	SC only	PPO

Per-Setting Generalization Curves

These per-w collapse-probability families underlie the cross-setting summary in Figure 3 (main text).

Formal Account: Mean-Field Criticality and the Calibration Gap

These two results (referenced from the Mechanism section) establish that the *magnitude* of w_{crit} is calibrated rather than predicted; the sign-invariance result that the central claim rests on (Theorem 1) is stated and proved in the main text.

Theorem 2 (Mean-field criticality, $N \to \infty$)**.** *If $\Sigma(x, N) = D(x)/N$, then as $N \to \infty$ the dynamics become deterministic, $\dot{x} = F(x, w, \lambda)$, and the coordinating basin disappears via a saddle-node bifurcation. The mean-field threshold $w_{\text{crit}}^{\text{MF}}$ is the unique solution of $F(x^*, w, \lambda) = 0$, $\partial_x F(x^*, w, \lambda) = 0$, and is a closed-form function of the payoff and prospect-theory parameters.*

Proof. With $\Sigma = D(x)/N \to 0$ the noise term in (10) vanishes and the flow is $\dot{x} = F$. A coordinating stable node

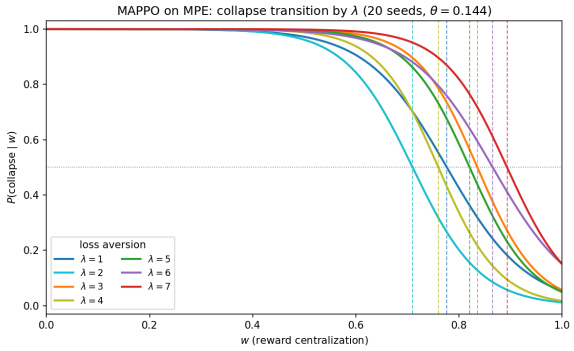


Figure 8: MAPPO on MPE Cooperative Navigation: fitted collapse-probability curves $P(\text{collapse} \mid w)$ for the full loss-aversion sweep λ=1, . . . , 7 (α=β=0.88, centralized critic, 50-step horizon, 20 seeds per cell; collapse = per-step landmark occupancy below θ=0.144, recalibrated from this run's null w=0/w=1 endpoints). The family shifts rightward as λ grows, with vertical dashed w_{crit} lines ordered accordingly—the same loss-aversion signature found in the supply chain, recovered under a different environment and a different (centralized-critic) algorithm.

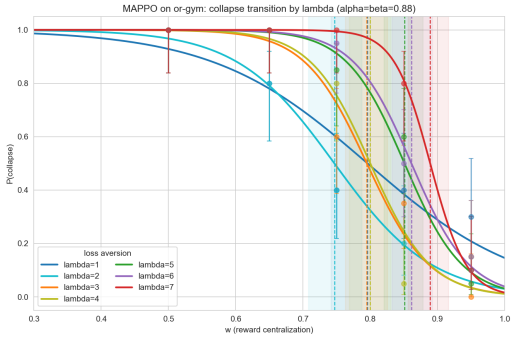


Figure 9: Centralized-critic MAPPO on the supply chain: fitted collapse-probability curves $P(\text{collapse} \mid w)$ for λ = 1, . . . , 7 (α=β=0.88, 300k steps, 20 seeds per cell, 700 runs). As under PPO, the family shifts rightward as loss aversion grows; the PPO-vs-MAPPO $w_{\text{crit}}(\lambda)$ comparison is shown alongside MPE in Figure 3.

x^*_{coord} and an unstable saddle x^*_{saddle} bound the basin; the basin vanishes when they collide, i.e. when $F = 0$ and $\partial_x F = 0$ coincide—the saddle-node condition. As F is closed-form in the payoff and CPT functions, so is its root $w^{\text{MF}}_{\text{crit}}$, which serves as the canonical baseline. □

Theorem 3 (Calibration gap, finite N and horizon T)**.** *For finite N with correlated (CTDE) updates, $\Sigma(x, N) > 0$, and collapse is a noise-induced escape rather than a bifurcation. The empirical boundary is the metastability root $\tau(w, \lambda) = T$ of the Kramers mean first-passage time*

$$\tau(w,\lambda) \asymp \exp\big(\Delta U(w,\lambda)\big), \qquad \Delta U = \int_{x^*_{\text{coord}}}^{x^*_{\text{saddle}}} \frac{-2F(z,w,\lambda)}{\Sigma(z,N)}\,\mathrm{d}z, \tag{12}$$

and $w^{\text{Emp}}_{\text{crit}} \neq w^{\text{MF}}_{\text{crit}}$ in general; the offset cannot be isolated without measuring Σ.

Proof. The stationary density of (10) is $P_{ss}(x) \propto \exp[-U_{\text{eff}}(x)]$ with $U_{\text{eff}}(x) = -\int_0^x 2F(z,w,\lambda)/\Sigma(z,N)\,\mathrm{d}z$, giving the barrier $\Delta U = U_{\text{eff}}(x^*_{\text{saddle}}) - U_{\text{eff}}(x^*_{\text{coord}})$ of (12), and Kramers' rate yields $\tau \asymp e^{\Delta U}$ (prefactor suppressed). At finite T, collapse occurs when $\tau \leq T$, so the boundary solves $\tau(w, \lambda) = T$. Since Σ is the (intractable) covariance of PPO updates, the value of ΔU—hence the root $w^{\text{Emp}}_{\text{crit}}$—requires empirical calibration. The two collapse notions coincide as limits: finite (N, T) escapes the basin *before* the Theorem 2 bifurcation, and $w^{\text{Emp}}_{\text{crit}}(T) \downarrow w^{\text{MF}}_{\text{crit}}$ as $T \to \infty$—the $\propto -\ln T$ slide observed above. □

Pre-registered Test of the Signal-to-Noise Mechanism

The mean-field account (Section) calibrates the magnitude of w_{crit} rather than predicting it, because the relevant drift and diffusion coefficients are properties of the stochastic learning dynamics, not static quantities of the cost matrix. We pre-register here a direct measurement that would upgrade the magnitude from calibrated to predicted, and we fix its analysis before running it.

Observable. The mean-field model treats the policy's mean base-stock action μ as a Langevin variable, $\mathrm{d}\mu = D(w)\,\mathrm{d}t + \sqrt{2\,\Theta(w)}\,\mathrm{d}W$. During PPO training we log μ once per update and, over a fixed early-training window, estimate the drift $D(w)$ (mean increment) and diffusion $\Theta(w)$ (increment variance / 2), forming $\text{SNR}(w) = D(w)/\sqrt{2\Theta(w)}$.

Prediction and decision rule. We sweep $w \in \{0.50, \ldots, 0.90\}$ at the structural condition (α=β=1, λ=1), 20 seeds per w, 150k steps, and compare against the *matched-horizon* boundary $w_{\text{crit}} = 0.822$ (Section). The mechanism is corroborated iff the measured signal-to-noise crossing's bootstrap 95% CI contains 0.822.

Outcome. The result is mixed, and we report it in full. The *collapse-fraction* of this independent sweep crosses 0.5 at $w = 0.82$, reproducing the matched-horizon boundary almost exactly—a positive robustness check on the boundary itself. The *drift-sign* observable, however, was *inconclusive*: the restoring drift in the contested band is weakly positive across all w with no genuine zero-crossing, so it does not localise w_{crit}. This is consistent with the risk-dominance reading of Section : the operative boundary is a *noise-bridged* crossing of the cooperative basin, not a sign flip of the deterministic drift $D(w)$, so the absence of a clean drift zero is expected rather than a failure of the mechanism. The reason is basin geometry: PPO initialises above the cooperative optimum (S≈250 vs S≈150), and collapses often plunge through the contested band too quickly to be sampled, biasing the conditioned drift toward survivors. We therefore do *not* claim a magnitude-level prediction; the boundary magnitude remains calibrated, as stated in Table 7. The mechanism section stands on its sign-level and functional-form predictions, which are unaffected. The driver and analysis (with the decision rule encoded) are released with the code so the test can be re-run with alternative observables (e.g. a longer horizon matched to $w_{\text{crit}} = 0.666$, or a basin-flow statistic).

Exploration-Noise Ablation

A further pre-registered experiment probes the boundary along an axis the main analysis holds fixed: the exploration noise of the learner. It reuses the identical substrate, PPO configuration, and collapse label, and provides the direct empirical support for the signal-to-noise mechanism (Section). (The heterogeneous-population result is promoted to the main text, App. .)

Exploration noise (entropy coefficient). The mean-field account (Section) predicts that w_{crit} is set by the signal-to-noise ratio of the policy gradient, so increasing the exploration term should raise it. We sweep the PPO entropy coefficient at the structural condition (α=β=1, λ=1, 300k, 20 seeds, $w \in \{0.5, \ldots, 0.9\}$). The boundary rises monotonically with exploration noise (Figure 10, Table 9): $w_{\text{crit}} = 0.685$ at ent=0 (reproducing the null baseline exactly), 0.690 at 0.001, 0.715 at 0.01, and beyond 0.05 the system collapses at *every* tested w (w_{crit} extrapolates above 1). This is direct empirical support for the signal-to-noise mechanism: the location of the boundary is not a function of the behavioral bias alone but also of the exploration noise that competes with the diluted cooperative signal, exactly as (8) predicts.

Table 9: Entropy-coefficient ablation (α=β=1, λ=1, 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs.

entropy coef.	w_{crit} (95% CI)
0.000	0.685 [0.656, 0.710]
0.001	0.690 [0.667, 0.715]
0.010	0.715 [0.689, 0.743]
0.050	> 1 (collapse at all w)

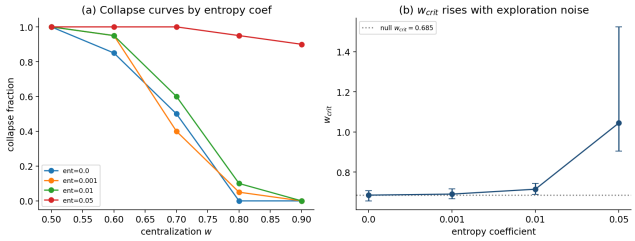


Figure 10: Exploration-noise ablation. (a) Collapse curves shift right as the entropy coefficient grows. (b) w_{crit} rises monotonically with exploration noise, corroborating the signal-to-noise mechanism (Section); ent=0 reproduces the null boundary 0.685.

Cross-environment: a binding-noise dissociation. We repeat the entropy sweep under centralized-critic MAPPO on the two transfer substrates (supporting the body claim of Section). On Level-Based Foraging the prediction *replicates*: w_{crit} rises monotonically with the entropy coefficient, $0.13 \to 0.15 \to 0.24 \to 0.37$ over ent $\in \{0, 0.001, 0.01, 0.05\}$ (shift $+0.24$, non-overlapping endpoint intervals). On MPE Cooperative Navigation the boundary does *not* move: across ent $\in \{0, \ldots, 0.12\}$ (extended past the seed-noise floor, with the policy verified non-degenerate at every level) the shift is $+0.05$ with overlapping intervals, and a pooled logistic regression over all 1,320 runs gives an entropy-coefficient slope positive in sign but not significant ($\beta{=}+0.47$, 95% CI $[-3.4, +4.3]$). This is a *dissociation predicted by the mechanism*: (8) says exploration noise moves the boundary only when it is the *binding* term in the signal-to-noise ratio. The supply chain and LBF have sparse, sharp reward landscapes in which the injected entropy dominates the gradient noise; MPE's categorical actions and dense, distance-shaped occupancy reward carry high *intrinsic* exploration that swamps the added term, so the boundary is insensitive to it—the effect appears exactly where the mechanism predicts and is absent exactly where it predicts.

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